

Demystifying 2D Time Series Forecasting: A Targeted Redesign of TimesNet for High-Frequency Data

Nicola Ranzolin and Matteo Gidoni[†]

Abstract—Short-to-medium term time series forecasting is critical for optimizing industrial and energy systems. Recent literature has explored 2D topological transformations, notably TimesNet, which utilizes Fast Fourier Transform (FFT) to dynamically extract multi-periodicities and process 1D signals as 2D spatial grids. While highly effective on long-term horizons, its period extraction introduces potential overhead with respect to simpler models in some scenarios due to tensor irregularities and struggles with spectral leakage on high-frequency, short-window datasets. In this project, we propose LightTimesNet, a targeted architectural redesign that replaces FFT-based dynamic extraction with domain-knowledge-driven static spatial patching. By enforcing a deterministic [frequency, period] topological reshape, LightTimesNet restores full hardware vectorization and eliminates batch-level iteration bottlenecks. We extensively evaluate our proposal against the original TimesNet and strong 1D baselines (DLinear, CausalTCN) on the ETTh1 and Electricity datasets for short-to-medium horizons ($H \in \{24, 48, 96\}$). Results demonstrate that injecting a priori domain knowledge into the 2D grid construction drastically reduces training overhead while maintaining competitive prediction accuracy, and that spatial backbone optimizations (e.g., Depthwise convolutions) yield a highly favorable Pareto front for edge deployments.

Index Terms—Time Series Forecasting, Convolutional Neural Networks, Spatial Patching, TimesNet, multi-horizon forecasting, periodicity modeling, lightweight neural networks, accuracy-efficiency trade-off.

I. INTRODUCTION

Time series forecasting is a fundamental task for cyber-physical systems, particularly in energy grid management where accurate predictions directly impact load balancing and cost efficiency. Traditionally, 1D architectural models have dominated this field, processing signals strictly along the temporal axis.

Despite their efficiency, pure 1D representations often fail to capture the complex inter-period and intra-period variations inherent in real-world data. Recent state-of-the-art approaches, such as TimesNet, attempt to solve this by transforming 1D series into 2D spaces based on multiple periodicities. However, the reliance on dynamic Fast Fourier Transform (FFT) to discover these periods hiddenly introduces significant computational overhead and memory fragmentation, limiting its deployment on resource-constrained hardware. Furthermore, its efficacy degrades on short lookback windows sampled at high frequencies.

In this work, we tackle the computational and representational inefficiencies of dynamic 2D transformations. We propose **LightTimesNet**, an optimized 2D architecture that abandons stochastic frequency extraction in favor of rigid, vectorized spatial patching driven by domain knowledge. This solution provides practitioners with a highly scalable architecture specifically tailored for short-to-medium term forecasting in environments with known seasonal behaviors (e.g., daily energy loads).

The main contributions of this report are as follows:

- We identify and critically analyze the topological and computational bottlenecks of TimesNet, specifically regarding hardware vectorization limits and spectral leakage on high-frequency datasets like Electricity.
- We design LightTimesNet, substituting the FFT module with a deterministic [frequency, period] tensor routing that enables $\mathcal{O}(1)$ batch-level spatial patching.
- We assess the benefits of 1D and 2D temporal representations across different sequence lengths.
- We investigate fixed-period modeling through a simplified TimesNet architecture.
- We analyze the impact of varying the top-k periods and the number of TimesBlocks.
- We evaluate alternative LightTimesNet backbones.
- We provide a final trade-off analysis in terms of accuracy, computational complexity, and inference speed.

II. RELATED WORK

Time-series forecasting methods can be broadly divided into efficient 1D architectures and models based on explicit 2D temporal representations.

a) Efficient 1D models.: DLinear showed that simple channel-independent linear mappings can outperform more complex architectures on several forecasting benchmarks [1]. Causal TCNs similarly provide efficient long-range modeling through dilated causal convolutions. These approaches are computationally attractive, but they process temporal patterns along a single axis and do not explicitly align observations belonging to the same periodic phase. In this work, they are used as strong baselines to assess whether 2D periodic representations provide additional benefits.

b) Two-dimensional temporal representations.: TimesNet introduced a 1D-to-2D transformation in which FFT-derived periods are used to reshape a sequence into intra-period and inter-period structures [2]. Multi-scale 2D Incep-

[†]Department of Information Engineering, University of Padua, email: {name.surname}@studenti.unipd.it

tion blocks then extract temporal patterns from each representation. Although effective, TimesNet combines adaptive period selection, multiple transformations, stacked TimesBlocks, and a complex backbone, making it difficult to isolate the contribution of each component.

Our work extends TimesNet through a controlled component-wise analysis. We evaluate fixed-period reshaping, the number of selected periods and TimesBlocks, and alternative lightweight backbones. These findings lead to LightTimesNet, which preserves 2D periodic modeling while improving the trade-off between forecasting accuracy, complexity, and inference speed.

III. PROCESSING PIPELINE

The proposed processing pipeline is designed to provide a fair comparison between 1D temporal models and 2D spatial representations for time-series forecasting. As illustrated in Fig. 1, the framework receives a multivariate historical sequence represented as $\mathbf{X} \in \mathbb{R}^{B \times T \times C}$, where B , T , and C denote the batch size, input sequence length, and number of variables, respectively.

The raw time-series data are first divided into training, validation, and test sets. A StandardScaler normalization is then applied, fitting the scaling parameters exclusively on the training split to prevent information leakage. The normalized data are converted into sliding windows, generating input-output pairs composed of historical observations and future prediction horizons:

$$\mathbf{X} \in \mathbb{R}^{B \times T \times C}, \quad \mathbf{Y} \in \mathbb{R}^{B \times H \times C},$$

where H represents the forecasting horizon.

The processed sequences are then provided to different forecasting architectures to analyze the impact of temporal and spatial representations. The evaluated approaches include 1D temporal baselines, such as linear and convolutional models, and 2D TimesNet-based variants. For the latter, the temporal sequence can be transformed into spatial representations through two alternative strategies: adaptive period extraction using FFT-based top- k frequency selection, or deterministic fixed-period reshaping.

In the original TimesNet formulation, dominant periods are extracted dynamically from the frequency domain and used to construct multiple 2D temporal planes. These representations are processed through TimesBlocks, which exploit 2D convolutions to capture intra-period and inter-period dependencies. Conversely, the proposed simplified variants replace the dynamic extraction mechanism with predefined periods or alternative lightweight backbones to reduce computational overhead.

Finally, the generated forecasts are evaluated using standard forecasting metrics, including Mean Squared Error (MSE) and Mean Absolute Error (MAE). All models share the same training loop, utilizing an Early Stopping mechanism monitoring the MSE to prevent overfitting. Learning curves are explicitly tracked to extract the true optimal weights $\mathcal{W}^*$ prior to performance degradation. The experimental analysis follows

a Change One Variable At a Time (COVAT) methodology, where individual architectural components are modified while keeping the remaining training conditions unchanged. While we acknowledge that architectural variables are not strictly orthogonal, our spatial kernels were fixed. Thus, isolating the window length T evaluated temporal context scalability rather than joint local minima. Consequently, COVAT served purely as a necessary computational trade-off against hardware-induced combinatorial explosions, enabling a focused analysis of forecasting accuracy, complexity, and velocity.

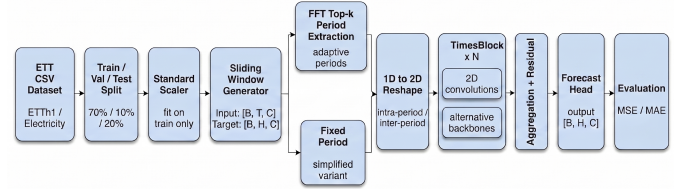


Fig. 1: Processing pipeline description.

IV. SIGNALS AND FEATURES

We conduct our evaluation on two standard multivariate benchmarks: the ETTh1 dataset (hourly sampling of 7 channels) and the Electricity dataset (hourly sampling of 321 individual clients). The inclusion of the Electricity dataset introduces high dimensionality and sparse activations, which severely test the spatial efficiency of 2D convolutions and the mathematical stability of forecasting metrics.

All experiments were implemented in PyTorch using the Metal Performance Shaders (MPS) backend and executed on an Apple M4 Pro SoC (12-core CPU, 16-core GPU, 24 GB unified memory).

Preprocessing and Split: The datasets are split chronologically into training (70%), validation (10%), and test (20%) sets to preserve temporal causality. Only the training windows are randomly shuffled during mini-batch generation, whereas validation and test samples retain their chronological order. Data is normalized using a standard scaler ($\mu = 0, \sigma = 1$). To prevent data leakage, the scaler is strictly fitted on the training split only, and subsequently applied to the validation and test sets. Regarding the Electricity dataset, it is imperative to specify that the aggregated hourly sampling is not an arbitrary downsampling procedure introduced in this study. This configuration is the official dataset provided publicly by the original authors of TimesNet in their repository. This methodological choice was executed explicitly to guarantee exact reproducibility and an honest 1:1 benchmarking against their proposed architecture, effectively eliminating any confounding variables linked to external sampling frequencies.

Windowing Strategy: The core of our evaluation is based on fixed-length sliding windows. We test sequence lengths $T \in \{96, 192, 384\}$ to evaluate temporal context scalability, predicting multiple short-to-medium term future horizons $H \in \{24, 48, 96\}$. Overlapping segments of length T from preceding sets are utilized purely as observational context

X for the earliest target windows Y , ensuring temporal continuity without target leakage.

V. LEARNING FRAMEWORK: LIGHTTIMESNET

This section describes the forecasting architectures evaluated in this work. The objective is to investigate the contribution of different temporal representations, period modeling strategies, and backbone designs while maintaining a consistent learning framework.

A. 1D Temporal Baselines

To establish a computationally efficient reference, two 1D temporal models are considered.

DLinear is adopted as a lightweight baseline that directly learns temporal dependencies through linear projections. By decomposing the input sequence into trend and residual components, DLinear provides a strong low-complexity reference for time-series forecasting. Its limited architectural complexity makes it suitable for evaluating the actual benefit introduced by more expressive representations.

CausalTCN extends the temporal modeling capability through causal one-dimensional convolutions. Unlike standard convolutions, causal convolutions ensure that each prediction only depends on past observations, preventing future information leakage. This architecture represents a convolution-based temporal routing strategy and provides a comparison between sequential and spatial feature extraction.

B. Original TimesNet

TimesNet introduces a different perspective by transforming one-dimensional temporal sequences into two-dimensional representations. Given an input sequence $X \in \mathbb{R}^{B \times T \times C}$, TimesNet first applies a Fast Fourier Transform (FFT) along the temporal dimension to identify dominant periodic components.

The selected frequencies are converted into corresponding periods and used to reshape the original sequence into multiple 2D temporal maps. Each map contains two complementary dimensions: the intra-period variation and the inter-period evolution. These representations are processed through TimesBlocks, which exploit two-dimensional convolutional operations to capture complex temporal patterns.

The original formulation dynamically selects the top- k dominant periods, allowing the model to adapt to different temporal behaviors. However, this adaptive mechanism introduces additional computational overhead due to dynamic period extraction, padding operations, and multiple 2D transformations.

C. TimesNet Variants

To analyze the contribution of individual architectural components, several TimesNet variants are investigated.

Top- k Period Selection. The number of selected dominant periods is varied to evaluate the effect of multi-period modeling. Configurations with different values of k are compared to understand whether additional periodic components improve forecasting performance or introduce redundant information.

TimesBlock Depth. The number of stacked TimesBlocks is modified to study the relationship between model capacity and forecasting accuracy. Increasing the number of blocks allows deeper feature extraction but also increases computational cost and the risk of over-parameterization.

Fixed-Period TimesNet. A simplified variant replaces the FFT-based dynamic period extraction with predefined temporal periods. This approach removes the adaptive period-selection stage and produces deterministic 2D transformations. The goal is to evaluate whether known domain periodicities can provide a more efficient alternative while maintaining competitive accuracy.

D. LightTimesNet Backbone

Based on the previous analysis, LightTimesNet adopts the fixed-period configuration identified as the most effective efficiency-accuracy compromise. The multi-block structure is further reduced to a single TimesBlock, limiting computational overhead while preserving the temporal-to-spatial representation paradigm.

Within LightTimesNet, different backbone configurations are investigated to analyze the impact of feature extraction complexity. The *Multiscale Inception* variant preserves the original multi-scale feature extraction principle of TimesNet by maintaining an Inception-inspired block with multiple convolutional kernels. Conversely, the remaining variants replace the original backbone with simplified operators, including *Group convolution*, *Single Kernel convolution*, and *Depthwise convolution*. These alternatives aim to improve computational efficiency by reducing the number of parameters and the cost of spatial feature extraction.

VI. RESULTS

The experimental analysis progressively investigates the components that led to the design of LightTimesNet. ETTh1 and Electricity were chronologically divided into 70% training, 10% validation, and 20% test samples, while normalization statistics were estimated exclusively from the training split. Only the training windows were randomly shuffled during mini-batch generation, whereas validation and test samples preserved their chronological order. MSE and MAE were used to evaluate forecasting accuracy. We started from Fixed-Period TimesNet, a controlled simplification of TimesNet that preserves its 2D temporal representation while removing adaptive period extraction.

A. Effect of Temporal Representation. 1D vs 2D: We first fixed the folding period to $P = 24$ and varied the input sequence length to determine whether a larger temporal context systematically benefits 2D processing. The comparison between the 1D baselines and the fixed-period 2D architecture is reported in Tab. 1. No consistent monotonic relationship between sequence length and forecasting accuracy was observed across all configurations. The clearest exception is the ETTh1 experiment with horizon $H = 48$, shown in Fig. 2, where the fixed-period model progressively reduces its MSE as the available context increases.

These results suggest that 2D representations benefit from longer input windows only when the additional context contains relevant and consistent periodic patterns. Therefore, increasing the sequence length alone does not guarantee improved forecasting accuracy. Since longer input windows did not provide a consistent advantage, we fixed the sequence length to 96 for the remaining experiments, reducing computational cost while preserving a representative temporal context.

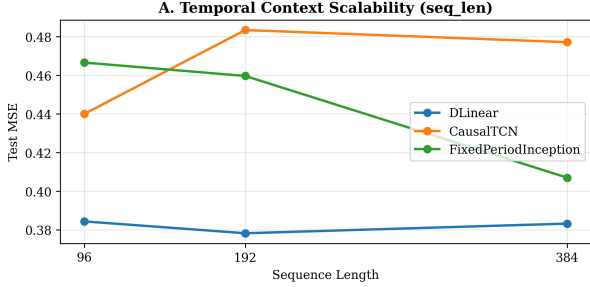


Fig. 2: Effect of the input sequence length on forecasting performance for ETTh1 Horizon 48, Period 24.

B. Fixed Period Analysis:

We then evaluated periods with different temporal meanings: $P = 24$, corresponding to a known daily cycle; $P = 48$, representing a multiple of this periodicity; and $P = 17$, which is not associated with an expected dataset pattern. The numerical results are reported in Tab. 2, while Fig. 3 illustrates the effect of the selected period on the forecasting error.

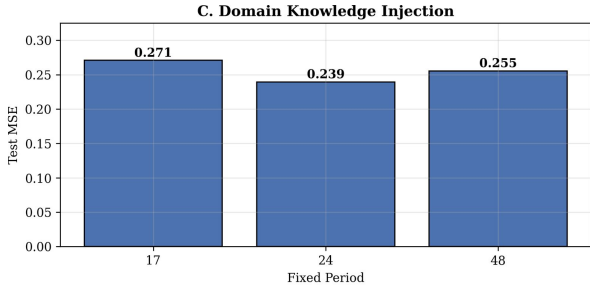


Fig. 3: MSE performance analysis for electricity, Horizon 24.

The known period and its multiple generally achieve lower errors than the non-informative alternative. This confirms that deterministic reshaping can be effective when reliable prior knowledge about the dataset periodicity is available. Consequently, $P = 24$ was retained in the subsequent LightTimesNet experiments.

C. Contribution of TimesBlocks and Multi-Period Modeling:

Tab. 3 compares the interaction between the number of dominant periods selected by TimesNet and the number of stacked TimesBlocks. As shown in Fig. 4, Fixed-Period TimesNet with a single TimesBlock remains consistently competitive while requiring fewer transformations and convolutional operations. The main exception is shown in Fig. 5,

where on the Electricity dataset ($H = 48$) the single-block fixed-period configuration incurs an MSE of 0.317, compared to 0.280 with three blocks: an error degradation of approximately 12%. We explicitly accept this accuracy penalty as a rigid architectural restriction. Stacking three blocks violates the latency and memory footprint ceilings required for lightweight edge deployments; thus, the single-block architecture is imposed as a mandatory hardware constraint rather than an optimal predictive minimum.

Moreover, increasing k in the original TimesNet does not produce a systematic reduction in MSE. In several configurations, the additional periodic representations increase computational cost without providing a proportional accuracy improvement. These observations motivated the main LightTimesNet design: deterministic period folding with $P = 24$ and a single TimesBlock.

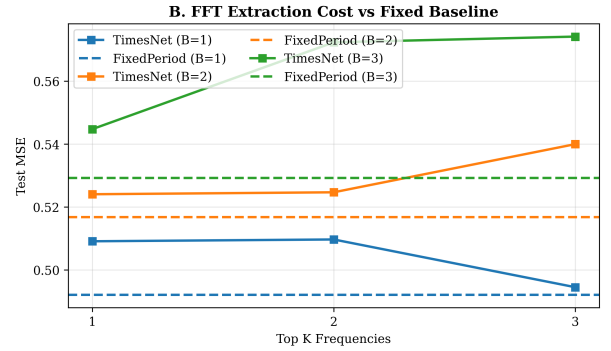


Fig. 4: Effect of FFT top- k selection and TimesBlock depth on test MSE, compared with fixed-period baselines. Measurements made on ETth1, Horizon 96

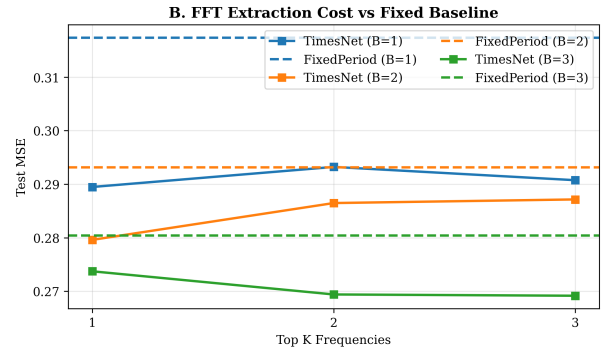


Fig. 5: Effect of FFT top- k selection and TimesBlock depth on test MSE. Exceptional MSE high value for fixed-period single TimesBlock. Measurements made on electricity, Horizon 48

D. Backbone Efficiency Analysis. Final Trade-off:

The results in Tab. 4 show that no backbone consistently dominates across both datasets. On ETTh1, as illustrated in Fig. 6, the Depthwise variant achieves the lowest MSE among all LightTimesNet configurations for every prediction horizon. It also uses approximately 71% fewer parameters and about 45% lower inference latency, on average, than the Multiscale

backbone. Furthermore, on ETTh1 at $H = 24$, the Depthwise variant uses fewer parameters than DLinear (4,191 vs. 4,656) while achieving a lower MSE (0.326 vs. 0.337). This result suggests that fixed-period spatial reshaping can provide a compact 2D representation that is competitive with, and in this configuration slightly outperforms, a channel-independent linear baseline without increasing model size.

A different behavior is observed on Electricity, as shown in Fig. 7. While the Multiscale backbone achieves the lowest overall MAE across the evaluated horizons, Group convolution provides a more favorable accuracy–complexity compromise than the Depthwise variant. At $H = 48$, Group convolution introduces approximately 2k additional parameters (28.9k vs. 26.9k) but reduces the MSE by about 14.4% (0.302 vs. 0.353).

This result suggests that partial cross-channel feature mixing can be beneficial on high-dimensional multivariate data. In contrast, the independent spatial filtering performed by Depthwise convolution may limit its ability to exploit dependencies among the Electricity variables. Therefore, Group convolution emerges as the stronger lightweight configuration for this setting, although additional experiments are required to establish whether the improvement is directly caused by cross-channel interactions.

Replacing TimesNet ($k = 1$, two TimesBlocks) configuration with the Group-based LightTimesNet variant on Electricity at $H = 48$ reduces the parameter count by approximately 83% (169k vs. 28.9k) and the total training time by about 60% (51.9s vs. 20.7s). This efficiency gain is obtained with a limited MSE increase of approximately 3.1% (0.293 vs. 0.302). These results indicate that the simplified architecture provides a favorable accuracy–efficiency trade-off.

The SingleKernel configuration does not provide a consistent advantage over the other simplified backbones.

The backbone analysis reveals a dataset-dependent trade-off rather than a universally optimal architecture. Depthwise convolution is the most effective LightTimesNet configuration on ETTh1, where additional multi-scale capacity does not improve forecasting accuracy. On Electricity, Multiscale provides the best LightTimesNet accuracy, while Group convolution achieves comparable performance with substantially fewer parameters. These findings suggest that Depthwise should be preferred when compactness is the main requirement, whereas Group offers a stronger compromise for higher-dimensional data. Nevertheless, the performance of DLinear demonstrates that lightweight 2D models should be considered efficient alternatives rather than universal replacements for strong 1D baselines.

VII. CONCLUDING REMARKS

This work provides a component-wise analysis of TimesNet and derives practical guidelines for building a lighter 2D forecasting architecture. The experiments show that fixed-period reshaping can replace adaptive FFT extraction when the dominant periodicity is known, while additional periods and TimesBlocks do not necessarily justify their computational cost.

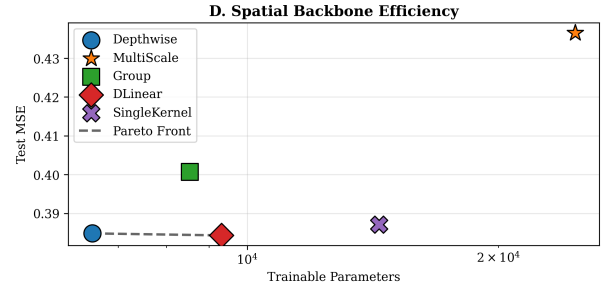


Fig. 6: Test MSE versus trainable parameters for the evaluated spatial backbones. Measurements made for Etth1, Horizon 48

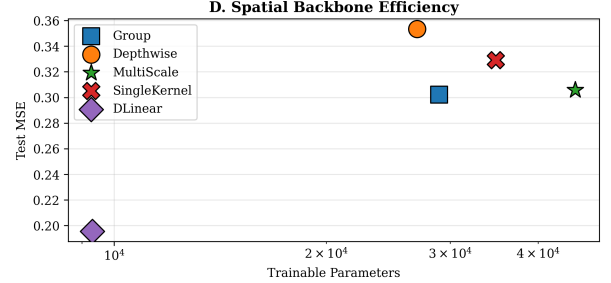


Fig. 7: Test MSE versus trainable parameters for the evaluated spatial backbones. Measurements made for electricity, Horizon 48

A central observation is that architectural complexity should reflect the complexity of the data. By enforcing hardware-aware constraints, such as capping the network to a single TimesBlock and adopting spatial Group convolutions for high-dimensional datasets, we demonstrated that a rigid 2D CNN can achieve up to an 83% parameter reduction and a 60% decrease in training latency compared to the original dynamic FFT baseline, while concurrently undercutting the parameter footprint of standard 1D linear mappings. Fixed-period reshaping and lightweight backbones are highly effective when dominant periodicities are stable, whereas adaptive period selection and multi-scale processing become a necessary computational trade-off only under irregular or time-varying patterns. Future work should therefore evaluate LightTimesNet on less regular datasets and investigate a hybrid mechanism that combines fixed-period efficiency with adaptive period selection. Additional COVAT ablations on cross-channel feature mixing and repeated runs across multiple seeds would further clarify the practical advantages of each backbone.

REFERENCES

- [1] A. Zeng, M. Chen, L. Zhang, and Q. Xu, “Are transformers effective for time series forecasting?,” in *AAAI Conference on Artificial Intelligence*, 2023.
- [2] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, and M. Long, “Timesnet: Temporal 2d-variation modeling for general time series analysis,” in *International Conference on Learning Representations*, 2023.

APPENDIX A

EXTENDED EXPERIMENTAL RESULTS

TABLE 1: Performance across different sequence lengths

		MAE									MSE								
		Seq 96			Seq 192			Seq 384			Seq 96			Seq 192			Seq 384		
Dataset	Pred	CT	DL	FPI	CT	DL	FPI	CT	DL	FPI	CT	DL	FPI	CT	DL	FPI	CT	DL	FPI
ETTh1	24	0.443	0.380	0.452	0.478	0.382	0.415	0.464	0.389	0.456	0.417	0.337	0.405	0.464	0.331	0.350	0.460	0.337	0.397
ETTh1	48	0.470	0.408	0.500	0.504	0.410	0.495	0.489	0.417	0.458	0.440	0.384	0.467	0.483	0.378	0.460	0.477	0.383	0.407
ETTh1	96	0.550	0.444	0.516	0.517	0.438	0.584	0.508	0.449	0.540	0.572	0.439	0.492	0.511	0.421	0.596	0.490	0.433	0.518
ETTh1	Avg	0.488	0.411	0.489	0.500	0.410	0.498	0.487	0.418	0.485	0.476	0.387	0.454	0.486	0.377	0.469	0.476	0.384	0.441
elec.	24	0.383	0.249	0.348	0.366	0.217	0.349	0.359	0.209	0.345	0.296	0.164	0.239	0.263	0.118	0.244	0.254	0.109	0.231
elec.	48	0.398	0.275	0.400	0.364	0.233	0.355	0.373	0.226	0.360	0.318	0.195	0.317	0.268	0.135	0.245	0.276	0.126	0.254
elec.	96	0.385	0.277	0.392	0.376	0.249	0.416	0.385	0.239	0.388	0.302	0.196	0.301	0.283	0.153	0.322	0.287	0.141	0.288
elec.	Avg	0.388	0.267	0.380	0.369	0.233	0.373	0.372	0.225	0.365	0.305	0.185	0.286	0.271	0.135	0.270	0.273	0.125	0.257

Note: CT = CausalTCN, DL = DLinear, FPI = FixedPeriodInception

TABLE 2: Period Sensitivity. Sequence length 96

		MAE					MSE				
Dataset	Pred	17	24	48	Avg	DLin	17	24	48	Avg	DLin
ETTh1	24	0.464	0.452	0.427	0.448	0.380	0.467	0.405	0.386	0.419	0.337
ETTh1	48	0.499	0.500	0.485	0.495	0.408	0.494	0.467	0.463	0.474	0.384
ETTh1	96	0.522	0.516	0.522	0.520	0.444	0.514	0.492	0.512	0.506	0.439
ETTh1	Avg	0.495	0.489	0.478	0.487	0.411	0.492	0.454	0.454	0.467	0.387
elec.	24	0.371	0.348	0.359	0.360	0.249	0.271	0.239	0.255	0.255	0.164
elec.	48	0.397	0.400	0.393	0.397	0.275	0.318	0.317	0.308	0.314	0.195
elec.	96	0.404	0.392	0.386	0.394	0.277	0.316	0.301	0.292	0.303	0.196
elec.	Avg	0.391	0.380	0.379	0.383	0.267	0.302	0.286	0.285	0.291	0.185

TABLE 3: Times Block vs Frequency (Top-K). Sequence length 96

			MAE									MSE								
			Pred 24			Pred 48			Pred 96			Pred 24			Pred 48			Pred 96		
Dataset	Model	Top-K	1B	2B	3B	1B	2B	3B	1B	2B	3B	1B	2B	3B	1B	2B	3B	1B	2B	3B
ETTh1	FPI	-	0.452	0.449	0.467	0.500	0.499	0.486	0.516	0.529	0.544	0.405	0.403	0.430	0.467	0.475	0.452	0.492	0.517	0.529
ETTh1	TN	1	0.454	0.453	0.460	0.504	0.491	0.480	0.528	0.525	0.552	0.400	0.404	0.415	0.465	0.462	0.442	0.509	0.524	0.545
ETTh1	TN	2	0.439	0.451	0.471	0.503	0.508	0.499	0.528	0.529	0.562	0.387	0.406	0.435	0.469	0.485	0.487	0.510	0.525	0.572
ETTh1	TN	3	0.451	0.472	0.461	0.483	0.492	0.498	0.518	0.540	0.561	0.401	0.436	0.417	0.451	0.464	0.491	0.495	0.540	0.574
elec.	FPI	-	0.348	0.367	0.346	0.400	0.389	0.370	0.392	0.404	0.380	0.239	0.269	0.240	0.317	0.293	0.280	0.301	0.315	0.287
elec.	TN	1	0.346	0.372	0.343	0.381	0.377	0.368	0.382	0.394	0.378	0.239	0.275	0.237	0.289	0.280	0.274	0.286	0.303	0.286
elec.	TN	2	0.345	0.382	0.345	0.384	0.383	0.362	0.384	0.396	0.378	0.237	0.287	0.239	0.293	0.286	0.269	0.288	0.310	0.287
elec.	TN	3	0.350	0.367	0.345	0.383	0.381	0.363	0.381	0.391	0.372	0.245	0.269	0.239	0.291	0.287	0.269	0.287	0.305	0.278

Note: 1B, 2B, 3B refer to Num Blocks. elec. = electricity, FPI = FixedPeriodInception, TN = TimesNet

TABLE 4: Backbone Efficiency Analysis. Sequence length 96

Dataset	Model	Pred	MSE	MAE	MASE	Params	Ckpt (MB)	Best Ep.	T/Best	Avg Ep.	Tot. T.	Inf (ms)
ETTh1	DLinear	24	0.337	0.380	1.550	4656	0.020	6	2.435	0.397	4.678	0.008
ETTh1	DLinear	48	0.384	0.408	1.724	9312	0.038	8	5.105	0.628	8.733	0.012
ETTh1	DLinear	96	0.439	0.444	1.939	18624	0.074	4	2.624	0.631	6.040	0.012
ETTh1	LTN_Depthwise	24	0.326	0.384	1.575	4191	0.020	5	4.812	0.967	10.146	0.009
ETTh1	LTN_Depthwise	48	0.385	0.421	1.779	6519	0.029	6	6.672	1.102	12.726	0.014
ETTh1	LTN_Depthwise	96	0.453	0.469	2.058	11175	0.047	5	5.719	1.122	11.795	0.015
ETTh1	LTN_Group	24	0.349	0.403	1.655	6207	0.028	4	4.202	1.045	9.909	0.010
ETTh1	LTN_Group	48	0.401	0.437	1.835	8535	0.037	8	9.323	1.159	15.811	0.014
ETTh1	LTN_Group	96	0.460	0.473	2.041	13191	0.055	1	1.465	1.298	8.166	0.014
ETTh1	LTN_MultiScale	24	0.356	0.411	1.670	22415	0.093	4	5.689	1.419	13.339	0.019
ETTh1	LTN_MultiScale	48	0.437	0.465	1.984	24743	0.101	6	10.217	1.689	19.385	0.022
ETTh1	LTN_MultiScale	96	0.479	0.496	2.292	29399	0.119	1	2.053	1.840	11.484	0.028
ETTh1	LTN_SingleKernel	24	0.343	0.399	1.623	12063	0.050	13	9.442	0.706	13.356	0.008
ETTh1	LTN_SingleKernel	48	0.387	0.430	1.834	14391	0.059	11	12.654	1.146	19.215	0.012
ETTh1	LTN_SingleKernel	96	0.478	0.488	2.141	19047	0.076	1	1.438	1.316	8.258	0.012
ETTh1	TN	24	0.387	0.439	1.851	146527	0.567	2	6.414	3.112	22.832	0.067
ETTh1	TN	48	0.469	0.503	2.257	148855	0.575	2	8.271	3.968	29.058	0.096
ETTh1	TN	96	0.510	0.528	2.371	153511	0.593	1	4.235	3.954	24.838	0.096
elec.	DLinear	24	0.164	0.249	0.988	4656	0.020	6	9.835	1.747	20.931	0.029
elec.	DLinear	48	0.195	0.275	1.103	9312	0.038	5	10.691	2.149	23.373	0.033
elec.	DLinear	96	0.196	0.277	1.116	18624	0.074	5	11.746	2.337	25.558	0.046
elec.	LTN_Depthwise	24	0.276	0.368	1.578	24601	0.098	14	28.876	2.056	41.718	0.030
elec.	LTN_Depthwise	48	0.353	0.424	1.869	26929	0.107	7	15.689	2.220	28.488	0.036
elec.	LTN_Depthwise	96	0.326	0.415	1.772	31585	0.125	4	10.599	2.632	25.350	0.043
elec.	LTN_Group	24	0.251	0.357	1.533	26617	0.106	4	8.440	2.085	20.065	0.029
elec.	LTN_Group	48	0.302	0.394	1.706	28945	0.115	3	7.404	2.429	20.712	0.039
elec.	LTN_Group	96	0.326	0.420	1.800	33601	0.133	5	13.176	2.628	28.137	0.046
elec.	LTN_MultiScale	24	0.244	0.352	1.530	42825	0.170	13	37.769	2.904	55.142	0.034
elec.	LTN_MultiScale	48	0.306	0.391	1.702	45153	0.179	20	45.508	2.275	48.894	0.038
elec.	LTN_MultiScale	96	0.317	0.401	1.738	49809	0.197	18	46.099	2.558	55.221	0.053
elec.	LTN_SingleKernel	24	0.252	0.355	1.538	32473	0.128	8	13.573	1.686	23.392	0.031
elec.	LTN_SingleKernel	48	0.329	0.409	1.776	34801	0.137	10	18.678	1.879	30.068	0.035
elec.	LTN_SingleKernel	96	0.331	0.420	1.826	39457	0.154	18	38.089	2.128	45.660	0.044
elec.	TN	24	0.237	0.345	1.501	166937	0.644	12	58.129	4.832	86.596	0.081
elec.	TN	48	0.293	0.384	1.659	169265	0.653	5	24.611	4.910	51.893	0.086
elec.	TN	96	0.288	0.384	1.648	173921	0.671	2	10.032	4.964	36.855	0.095

Note: LTN = LightTimesNet, TN = TimesNet (K = 2, TimesBlocks = 1)